

BYRON BAY, NU, Canada

WMO: 719290

Lat: 68.750N

Lon: 109.070W

Elev: 92

StdP: 100.22

Time Zone: -7.00 (NAM)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.5	-40.2	-39.7	0.1	-36.0	-39.1	0.1	-35.8	N/A	N/A	N/A	N/A	N/A	N/A	1.113

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.8	17.2	12.5	15.4	11.4	13.7	10.5	13.1	16.5	11.8	14.8	10.6	13.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	8.4	15.3	9.8	7.6	13.6	8.6	7.0	12.4	37.3	16.7	33.8	14.9	30.9	13.5	18.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-43.2	20.9	1.3	2.0	-44.2	22.4	-44.9	23.6	-45.6	24.7	-46.5	26.2
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.6	-32.3	-31.6	-27.9	-20.4	-8.9	3.3	9.5	6.6	0.1	-10.4	-23.8	-28.4
	DBStd	15.90	5.41	6.20	5.53	6.03	5.74	4.78	3.26	3.58	3.74	5.99	6.40	5.61
	HDD10.0	8663	1312	1164	1176	912	586	207	51	119	298	632	1016	1191
	HDD18.3	11649	1571	1397	1434	1162	844	450	274	364	548	891	1266	1449
	CDD10.0	56	0	0	0	0	0	7	36	13	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	120	3	3	3	4	6	10	25	26	18	14	5	3
	PrecMax	203	33	19	21	16	22	76	70	60	78	29	11	29
	PrecMin	69	0	0	0	0	0	0	2	2	2	3	1	0
	PrecStd	39	6	5	4	4	6	14	18	14	16	7	3	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.0	-11.2	-10.6	-3.4	3.2	17.6	21.2	18.7	9.7	3.2	-4.8	-9.2
		MCWB	-12.3	-11.5	-11.1	-4.0	2.3	11.4	15.3	13.6	7.8	2.9	-5.1	-9.4
	2%	DB	-16.6	-16.9	-14.1	-6.4	1.3	14.8	18.1	16.2	7.8	0.8	-7.8	-13.6
		MCWB	-17.0	-17.4	-14.6	-6.9	0.9	10.4	12.8	12.6	6.8	0.5	-8.1	-14.0
	5%	DB	-20.5	-19.5	-16.9	-8.7	0.2	12.4	16.4	14.0	6.4	-0.7	-10.9	-16.1
		MCWB	-20.9	-19.9	-17.3	-9.1	-0.1	9.3	11.9	11.4	5.5	-0.9	-11.3	-16.4
	10%	DB	-24.7	-22.9	-19.7	-11.5	-1.0	10.2	14.8	12.0	4.8	-2.2	-13.9	-19.7
		MCWB	-24.9	-23.1	-20.0	-11.9	-1.2	7.7	10.9	9.5	4.1	-2.5	-14.3	-20.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.8	-11.1	-10.8	-3.9	2.6	12.2	15.9	14.5	8.2	3.1	-5.0	-9.3
		MCDB	-11.5	-10.8	-10.4	-3.5	3.2	15.7	20.2	18.0	9.0	3.3	-4.8	-9.0
	2%	WB	-16.0	-14.5	-14.2	-6.8	0.9	10.7	13.7	12.8	6.7	0.4	-8.2	-13.9
		MCDB	-15.7	-14.2	-13.9	-6.5	1.2	14.2	17.2	15.7	7.6	0.7	-7.8	-13.6
	5%	WB	-18.6	-18.4	-16.9	-9.1	0.0	9.3	12.3	11.5	5.6	-0.9	-11.3	-16.2
		MCDB	-18.3	-18.1	-16.5	-8.7	0.2	12.6	15.7	13.9	6.3	-0.7	-10.9	-15.9
	10%	WB	-22.8	-20.8	-19.5	-11.8	-1.2	7.7	11.2	9.8	4.1	-2.5	-14.3	-19.6
		MCDB	-22.5	-20.4	-19.2	-11.5	-0.9	10.1	14.2	11.7	4.8	-2.2	-13.9	-19.4
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.3	5.5	6.3	5.1	4.7	5.8	4.8	3.4	4.8	4.9	5.0
		MCDBR	9.5	8.2	7.8	7.3	4.5	7.5	8.2	7.5	4.5	4.4	5.7	7.1
		MCWBR	9.4	7.7	7.7	7.2	4.4	5.2	4.9	4.9	3.9	4.4	5.7	7.1
	5% WB	MCDBR	9.5	8.1	7.9	7.3	4.5	7.5	7.9	7.4	4.4	4.3	5.8	7.2
		MCWBR	9.4	7.9	7.8	7.2	4.5	5.4	5.5	5.2	4.1	4.4	5.7	7.2
Clear-Sky Solar Irradiance	taub	0.146	0.206	0.242	0.289	0.318	0.343	0.356	0.335	0.292	0.230	0.158	N/A	
	taud	1.901	2.073	2.227	2.142	2.144	2.312	2.386	2.475	2.505	2.193	1.901	N/A	
	Ebn at Noon	105	597	808	852	863	852	827	808	758	550	67	0	
	Edh at Noon	5	43	72	107	121	107	97	79	58	37	5	0	
All-Sky Solar Radiation	RadAvg	0.02	0.45	2.08	4.29	6.34	6.30	5.40	3.50	1.67	0.53	0.07	0.00	
	RadStd	0.00	0.05	0.16	0.30	0.50	0.34	0.37	0.31	0.19	0.08	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page